
Random Subset Sampling for Multitrait GWAS: A Scalable Alternative to Exhaustive Phenotype Testing

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Abstract

Genome-wide association studies (GWAS) test one SNP against one phenotype at a time, which can miss a moderate, coordinated effect spread across correlated phenotypes; testing all phenotypes jointly avoids this but is combinatorially intractable and dilutes signal carried by only a subset of them. We implement and validate a random subset sampling strategy that estimates joint association from $N_s = 10,000$ randomly drawn phenotype subsets rather than the full combinatorial space, applied at full scale to 9,611,261 SNPs across 1,010 UK Biobank brain MRI phenotypes. The method finds 666 SNPs that are significant under subset-based testing but not the full joint test. Ranking phenotypes by score and testing them cumulatively, calibrated against a matched-size simulated null, identifies 173,033 SNPs whose joint signal cannot be attributed to the ranking procedure, and cross-validates the subset-based finding directly: 529 of these are significant only under this constructed subset, a comparable order of magnitude to the 666 found by random sampling despite the two constructions drawing from almost entirely non-overlapping regions of the combinatorial space. More broadly, the constructed subset typically matches or exceeds full joint-test power using a median of only 94 of the 1,010 phenotypes, showing that a SNP’s multitrait signal is often concentrated in, and recoverable from, a small and identifiable subset rather than the full phenotype pool.

1 Introduction

A GWAS tests, SNP by SNP, whether carrying an effect allele is associated with a measured phenotype across a cohort [8]. For a single phenotype k , the estimated effect of SNP j is summarized by the z-score $z_{j,k} = \hat{\beta}_{j,k}/SE(\hat{\beta}_{j,k})$, which is standard normal under the null hypothesis of no association. Running this test across M SNPs and K phenotypes produces a z-score matrix $\mathbf{Z} \in \mathbb{R}^{M \times K}$, one row per SNP and one column per phenotype. The work presented here operates only on this matrix of GWAS summary statistics, never on individual-level genotype or phenotype data.

The phenotypes analyzed in this article are $K = 1,010$ structural measurements derived from brain MRI scans of a UK Biobank (UKB) cohort of 32,947 individuals [3], tested against $M \approx 9.6$ million SNPs. The phenotypes fall into three categories: cortical surface area, cortical thickness, and subcortical or whole-brain volumes. Unlike a single disease outcome, these 1,010 phenotypes are not independent measurements. They are all derived from the same scan of the same individual, and many are anatomically and developmentally related, so substantial correlation between phenotypes is expected and must be accounted for statistically.

The GWAS model above is univariate: it tests one SNP against one phenotype at a time. Applied across all 1,010 phenotypes, this means running 1,010 separate tests per SNP, each blind to the fact that many of these phenotypes are correlated and may share a common underlying genetic signal. A SNP with a moderate, coordinated effect across many related brain structures may fail to reach significance in any single univariate test, yet be strongly supported when the phenotypes are considered jointly. Multitrait analysis addresses this by testing the association between a SNP and several phenotypes simultaneously, accounting for their correlation, which can substantially increase statistical power for SNPs with such coordinated effects.

The cost of this approach is combinatorial. From a pool of K phenotypes there is no single, unique way to group phenotypes for a joint test, and the number of possible non-empty subsets is $2^K - 1$. For $K = 100$ there are already over 10^{30} possible subsets, and for $K = 1,010$ approximately 10^{304} , making exhaustive search infeasible. Testing all 1,010 phenotypes together in a single joint test avoids the combinatorial problem but reintroduces the dilution it was meant to solve: a signal carried by a small group of related structures can be averaged away once all 1,010 traits are included. What is needed is a way to identify which subsets of phenotypes carry the largest association signal for a given SNP, without enumerating them all.

This article implements and validates a random subset sampling strategy for this problem (Figure 1), and scales it to the full brain MRI dataset. Rather than enumerating all possible subsets, N_s random subsets of size k are drawn from the pool of K phenotypes, and each is tested jointly using the omnibus statistic $\mathbf{S} = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z}$, where $\hat{\Sigma}_k$ is the null covariance matrix of the selected phenotypes. Statistics from subsets of different sizes are then rescaled onto a common χ_1^2 scale so that they remain comparable. The sampling strategy itself was addressed in an ongoing project at the Statistical Genetics unit at Institut Pasteur (Paris, France), which demonstrates that the otherwise intractable problem of selecting which phenotypes to test jointly can be resolved through random sampling. The contribution here is its implementation, verification, and application at full genome-wide scale to a phenotype pool that is both large and heavily correlated with the development of new methods for downstream analysis.

Specifically, this article contributes the following:

1. A **validated implementation of random subset sampling** that runs at full scale, over 9,611,261 SNPs and all 1,010 phenotypes, using the LDSC intercept matrix [2] as the null covariance estimator. Correctness was checked against a naive reference implementation at each stage, and two numerical issues (float32 underflow in the rescaling step, and an accumulation bug in the batched einsum) were identified and fixed before the method was trusted at the next stage.
2. A **phenotype score** that attributes a subset’s joint signal back to individual phenotypes by averaging the rescaled statistic over only the subsets each phenotype was drawn into, and a demonstration that this score surfaces phenotypes with elevated multitrait signal despite a weak or ordinary univariate z-score.
3. A **cumulative joint testing** procedure that ranks phenotypes by score and tests nested prefixes to construct each SNP’s peak-significance subset, calibrated against a genome-wide null to identify 173,033 SNPs whose signal exceeds selection bias alone, with a systematic eigenvalue scan ruling out near-singular covariance as a broad driver (26 SNPs flagged).
4. An **empirical comparison of univariate, full joint, and subset-based testing** across all 9,611,261 SNPs, quantifying agreement between the three and identifying 666 SNPs that are significant under subset-based testing but not the full joint test.
5. A **cross-validation of this finding** via the score-ranked subset in place of a random one, recovering a comparable 529 joint-invisible SNPs despite drawing from a largely non-

overlapping region of the combinatorial space, and more broadly, matching or exceeding full joint-test power using a median of only 94 of the 1,010 phenotypes.

Section 2 describes the data, the null covariance estimator, and the sampling pipeline. Section 3 presents the validation and the full-scale results. Section 4 discusses what the method does and does not provide.

2 Methods

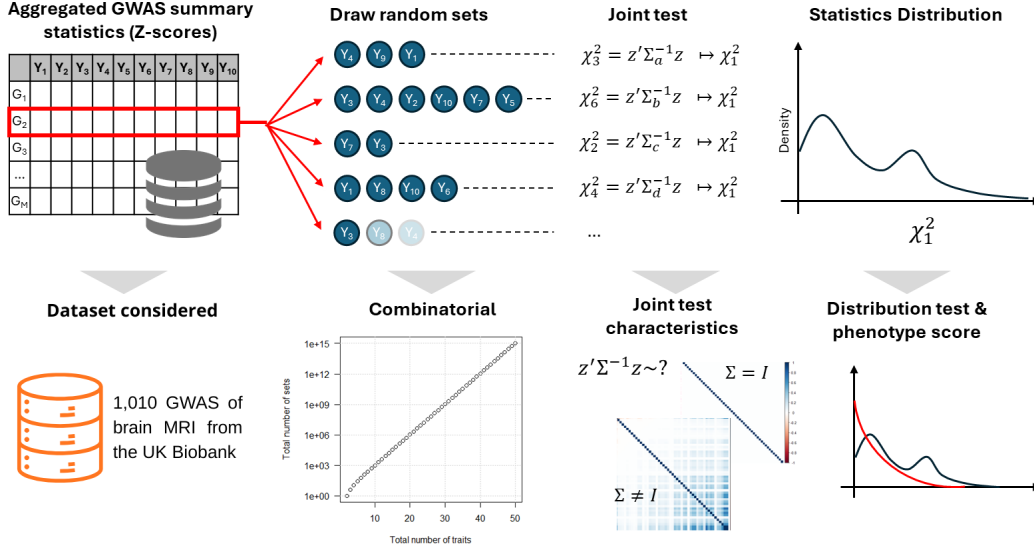


Figure 1: Pipeline illustration

2.1 Dataset

Most genetic variation between individuals takes the form of Single Nucleotide Polymorphisms (SNPs): single positions in the genome at which individuals carry different bases. Each SNP has two possible alleles, one of which is designated the effect allele, and an individual’s genotype at that SNP is encoded as the number of copies of the effect allele they carry: 0, 1, or 2. Nearby SNPs on the same chromosome tend to be co-inherited and are therefore correlated, a property called linkage disequilibrium (LD). This becomes relevant in Section 2.2 when estimating the null covariance matrix.

GWAS had already been performed across all 1,010 brain MRI phenotypes for this UKB cohort, producing a z-score matrix $\mathbf{Z} \in \mathbb{R}^{M \times K}$ of shape approximately 9.6 million $\times$ 1,010, with entry $z_{j,k} = \hat{\beta}_{j,k} / \text{SE}(\hat{\beta}_{j,k})$. The regression producing these values happens once, upstream; the pipeline (Figure 1) operates only on its output.

The raw table (`all_z_scores.txt` $\approx$ 82 GB) was streamed via Polars lazy scanning directly into a Parquet file, with phenotype columns reordered to match the LDSC intercept matrix for consistent submatrix indexing, and cast to `float32`. The conversion was verified by comparing the first 10,000 rows cell-by-cell against the original file: column order, SNP identifiers, and z-scores all matched exactly.

Some entries of the matrix are missing. Rows containing any missing phenotype value were removed rather than zero-filled, since zero-filling would bias the joint test toward the null. Of the 9,611,405 SNPs, 144 (0.0015%) contained at least one NaN z-score and were dropped, leaving a final working matrix of shape 9,611,261 $\times$ 1,010. This was written to disk as `Z_matrix.npy` and loaded via memory-mapping for the sampling procedure.

2.2 Null covariance estimator

The omnibus statistic defined in Section 1 requires a covariance matrix $\hat{\Sigma}_K$ that reflects the correlation structure of the z-scores under the null hypothesis, that is, the covariance arising solely from sample overlap and phenotypic correlation, with no genetic signal. Three estimators were considered.

The empirical phenotypic correlation, computed directly from the columns of the z-score matrix, conflates null and causal SNPs. Shared genetic effects inflate the off-diagonal entries, causing $\hat{\Sigma}_K^{-1}$ to be under-estimated and the test statistic to be deflated, which yields conservative and underpowered tests. A second estimator, inspired by JASS [6], restricts the correlation estimate to SNPs with weak marginal signal ($|z| < 1.96$ for every phenotype), on the reasoning that such SNPs are unlikely to carry true genetic effects, so their z-score correlation reflects only sample overlap and phenotypic correlation. This is a computationally accessible proxy for the correct estimator, but it is not statistically equivalent to it.

The estimator used throughout this article is the LDSC intercept matrix [2]. LD score regression regresses the product $z_{j,i} z_{j,k}$ on the LD score ℓ_j across all SNPs, exploiting the decomposition:

$$\mathbb{E}[z_{j,i} z_{j,k}] = \ell_j [\Sigma_g]_{i,k} + [\Sigma_{\text{int}}]_{i,k},$$

where the slope $[\Sigma_g]_{i,k}$ absorbs all LD-mediated genetic covariance and the intercept $[\Sigma_{\text{int}}]_{i,k}$ isolates the null covariance due to sample overlap and phenotypic correlation. The intercept is therefore uncontaminated by shared genetic effects, and is an unbiased estimator of the null covariance matrix. The full derivation is given in Appendix A.

Since all 1,010 phenotypes are measured on the same UKB cohort, every pair of traits shares all individuals, so the sample-overlap coefficient reduces to one and the intercept collapses to:

$$[\Sigma_{\text{int}}]_{i,k} = \frac{N^{(i,k)}_{\text{overlap}} \rho_{i,k}}{\sqrt{N_i N_k}} = \rho_{i,k},$$

the phenotypic correlation between traits i and k . The matrix $\hat{\Sigma}_{1010}$ used here is therefore the $1,010 \times 1,010$ LDSC intercept correlation matrix (Figure 2), which had been pre-computed for these phenotypes and required only reordering to match the phenotype columns of the z-score matrix. It was cast to `float32` to match the precision of the sampling loop.

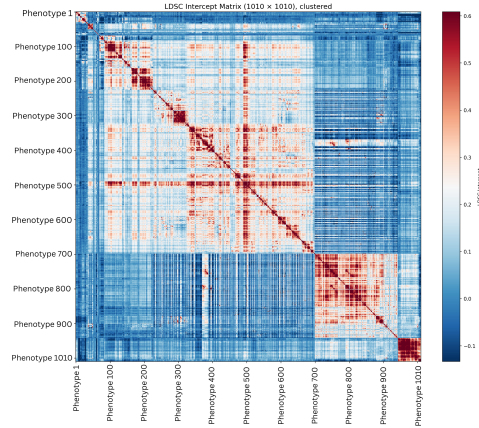


Figure 2

The Cholesky factorization used in Section 2.3.1 requires $\hat{\Sigma}_k$ to be positive definite. This was confirmed explicitly on smaller blocks by checking that all eigenvalues are strictly positive: the minimum eigenvalue was 0.225473 for the 100×100 block and approximately 0.012 for the 500×500 block. At full scale each subset's $k \times k$ submatrix is Cholesky-factorized directly at the point of use, which fails immediately if the matrix is not positive definite, so the successful completion of all 10,000 factorizations serves as an implicit confirmation.

2.3 Sampling pipeline

The remainder of this section breaks down the pipeline illustrated in Figure 1 and Algorithm 1. Subsets are drawn at random from the phenotype pool and each is tested jointly against every SNP, the resulting statistics are rescaled onto a common scale so that subsets of different sizes remain comparable, and the rescaled statistics are then averaged per phenotype to give a phenotype score. The dataset of Section 2.1 and the null covariance estimator of Section 2.2 are the two inputs this pipeline requires.

Algorithm 1 Fullscale Random Subset Sampling

Require: $\mathbf{Z} \in \mathbb{R}^{N_{\text{SNP}} \times N_p}$, $\hat{\Sigma} \in \mathbb{R}^{N_p \times N_p}$, $N_s = 10,000$, $k_{\min} = 5$, $k_{\max} = 90$
Ensure: $\mathbf{T} \in \mathbb{R}^{N_s \times N_{\text{SNP}}}$ (rescaled statistics), $\mathbf{A} \in \{0, 1\}^{N_s \times N_p}$ (selection indicators)

- 1: **for** $i = 1, \dots, N_s$ **do**
- 2: Draw $k_i \sim \text{Uniform}\{k_{\min}, \dots, k_{\max}\}$
- 3: Draw $\mathcal{S}_i \subseteq \{1, \dots, N_p\}$ without replacement, $|\mathcal{S}_i| = k_i$
- 4: Compute Cholesky factorization $\mathbf{L}_i \mathbf{L}_i^\top = \Sigma_{\mathcal{S}_i}$
- 5: **for** $j = 1, \dots, N_{\text{SNP}}$ **do**
- 6: Solve $\mathbf{w} = \mathbf{L}_i^{-1} \mathbf{z}_{j, \mathcal{S}_i}$ (forward substitution)
- 7: $S_{i,j} \leftarrow \mathbf{w}^\top \mathbf{w} = \mathbf{z}_{j, \mathcal{S}_i}^\top \Sigma_{\mathcal{S}_i}^{-1} \mathbf{z}_{j, \mathcal{S}_i} \sim \chi_{k_i}^2$
- 8: $p_{i,j} \leftarrow \Pr(\chi_{k_i}^2 > S_{i,j})$
- 9: $T_{i,j} \leftarrow F_{\chi_1^2}^{-1}(1 - p_{i,j})$ (rescale to χ_1^2)
- 10: **end for**
- 11: $\mathbf{T}[i, :] \leftarrow T_{i, :}$
- 12: $\mathbf{A}[i, \mathcal{S}_i] \leftarrow 1$
- 13: **end for**

2.3.1 Random subset sampling and the omnibus test

Rather than enumerating all possible subsets for K phenotypes, N_s random subsets are drawn from the pool. For each subset $i \in \{1, \dots, N_s\}$, a size k_i is drawn uniformly from a fixed range and k_i phenotypes are then selected without replacement from the K columns of the z-score matrix. At full scale, $N_s = 10,000$ subsets were drawn with k_i sampled uniformly from $\{5, \dots, 90\}$ and phenotypes drawn from the full pool of 1,010. The lower bound of 5 is set to avoid trivially small subsets. Reproducibility was ensured via a NumPy `SeedSequence` spawning independent child seeds, one per chunk of subsets.

For a subset of k phenotypes, the joint test asks whether the SNP has an effect on any phenotype in that subset. Writing $\beta = (\beta_1, \dots, \beta_k)^\top$ for the effect of the SNP across the k selected phenotypes, the hypotheses are

$$H_0 : \beta = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0 \text{ for at least one } j,$$

compared to the univariate case, where $H_0 : \beta = 0$ is tested against $H_1 : \beta \neq 0$ for a single phenotype. The alternative here is therefore composite: rejecting H_0 establishes that the subset carries signal, but does not identify which of the k phenotypes is responsible. This is what makes attributing signal back to individual phenotypes difficult, and it is the reason the subset-membership matrix is retained alongside the statistics and the phenotype score of Section 2.3.3 is computed.

For a subset with z-score vector $\mathbf{z} \in \mathbb{R}^k$, the test statistic is the Mahalanobis distance [7] :

$$S = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \Sigma_k) \text{ under } H_0 \implies S \sim \chi_k^2,$$

where $\hat{\Sigma}_k$ is the $k \times k$ submatrix of the LDSC intercept matrix restricted to the selected phenotypes. Two simplifying assumptions hold throughout: $\hat{\Sigma}_k$ is treated as identical for every SNP, and each z-score is standard normal under the null hypothesis.

Rather than forming $\hat{\Sigma}_k^{-1}$ explicitly at $O(k^3)$ cost, $\hat{\Sigma}_k$ is Cholesky-factorised as $\hat{\Sigma}_k = LL^\top$ [5], and the triangular system $LY = \mathbf{z}$ is solved for $Y = L^{-1}\mathbf{z}$ at $O(k^2)$ cost. Since $Y \sim \mathcal{N}(0, I_k)$ under H_0 , the statistic reduces to a sum of squares:

$$S = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z} = \|Y\|^2 \sim \chi_k^2.$$

Because L depends only on the selected phenotype subset and not on the SNP, it is computed once per subset and reused across all SNPs. Arranging every SNP as a column gives a single batched triangular solve, one BLAS call, that produces Y for every SNP simultaneously. This reduces the

per-subset cost from $O(k^3 \times N_{\text{SNP}})$ to one $O(k^3)$ factorization plus one $O(k^2 \times N_{\text{SNP}})$ batched solve, and is what makes the procedure tractable at genome-wide scale.

The procedure yields two matrices, both indexed by subset along the first axis. The first, `stat_matrix` $\in \mathbb{R}^{N_s \times N_{\text{SNP}}}$, holds the association statistic for each subset and SNP, rescaled as described in Section 2.3.2. Entry (n, j) is this value for SNP j under subset n , with larger values indicating stronger evidence that the phenotypes in subset n , taken jointly, are associated with SNP j . Since $N_s \times N_{\text{SNP}}$ reaches approximately 9.6×10^{10} entries at full scale, this matrix is stored at `float32` precision as a memory-mapped file rather than held in RAM. The second, `sel_matrix` $\in \{0, 1\}^{N_s \times K}$, is the subset-membership matrix, with entry $(i, p) = 1$ if phenotype p was drawn in subset i and 0 otherwise, stored as `uint8`. Each row therefore contains exactly k_i ones.

Together these two matrices are sufficient to reconstruct every subsequent analysis without rerunning the sampling loop: `stat_matrix` supplies the per-SNP, per-subset association strength, and `sel_matrix` supplies the phenotype membership needed to attribute that strength back to individual phenotypes.

2.3.2 Rescaling to a common scale

Subsets of different sizes k produce χ_k^2 statistics on different scales, which are not directly comparable. Each statistic is therefore rescaled to its 1-degree-of-freedom equivalent via the p-value-preserving transform

$$S_1 = F_{\chi_1^2}^{-1} \left(F_{\chi_k^2}(S) \right),$$

which satisfies $P(\chi_1^2 > S_1) = P(\chi_k^2 > S) = p$ by construction. The two statistics therefore carry the same tail probability, now expressed on a common scale regardless of k . The survival and inverse-survival functions are supplied by `chdtrc` and `chdtri` from `scipy.special`, which are faster than their `scipy.stats` equivalents as they bypass the `chi2` object-wrapper overhead. Appendix C gives the derivation of both functions and shows that the transform preserves tail probabilities exactly.

Two numerical precautions are required. First, the statistic is cast from `float32` to `float64` before the survival function is evaluated: for strongly associated SNPs the tail probability $P(\chi_k^2 > S)$ can fall below 10^{-38} , which underflows to exactly 0.0 in `float32` and propagates `inf` into the rescaled output. Second, p-values are clipped to $[10^{-300}, 1]$ before the inverse transformation, so that the transform remains numerically stable for the most extreme statistics. The resulting `float64` value is cast back to `float32` on assignment into `stat_matrix`.

The transformation is applied inside the sampling loop immediately after each subset's statistics are computed, operating on the vector of statistics for that subset across all SNPs rather than on the full $N_s \times N_{\text{SNP}}$ matrix at once. This avoids materialising the entire rescaled matrix in RAM. At full scale, approximately 9.6×10^{10} statistics were rescaled across all 10,000 subsets.

2.3.3 Phenotype score

The sampling procedure yields, for every SNP, up to N_s estimates of association strength, one per randomly drawn subset, each already rescaled onto the common χ_1^2 scale. To summarise this information per phenotype rather than per subset, a phenotype score is computed: for a given SNP and a given phenotype, the average of the rescaled statistic over only those subsets that included that phenotype. Writing `sel` $\in \{0, 1\}^{N_s \times N_p}$ for the subset-selection matrix and `stat` $\in \mathbb{R}^{N_s \times N_{\text{SNP}}}$ for the matrix of rescaled statistics, the score matrix $\in \mathbb{R}^{N_{\text{SNP}} \times N_p}$ is:

$$\text{score}[j, p] = \frac{\sum_{n=1}^{N_s} \text{stat}[n, j] \cdot \text{sel}[n, p]}{\sum_{n=1}^{N_s} \text{sel}[n, p]},$$

the mean rescaled statistic at SNP j , averaged only over the subsets in which phenotype p participated. A phenotype scoring highly here consistently contributes strong multitrait signal across every subset it was drawn into, identifying traits that are reliably associated with a SNP regardless of which other phenotypes happened to accompany them in a given subset.

The denominator, $\text{counts}[p] = \sum_n \text{sel}[n, p]$, depends only on the sampling scheme and is computed once. The numerator, for a block of SNPs, is a single matrix product $\text{stat}^\top \text{sel}$, exploiting the fact that multiplying by the binary selection matrix and dividing by the per-phenotype counts reduces the sum to only the selected subsets. The score computation is therefore a single pass of dense matrix multiplication over already-computed statistics, with no per-subset factorization or survival function evaluation.

2.4 Cumulative joint testing

The omnibus test establishes that a subset carries signal but does not identify which phenotypes within it are responsible (Section 2.3.1). To construct the subset directly rather than infer it from subset membership, the phenotypes are ranked and tested cumulatively.

For a given SNP, the 1,010 phenotypes are sorted in descending order of their phenotype score at that SNP, giving an ordering $p_{(1)}, \dots, p_{(1010)}$. The joint statistic is then computed on each nested prefix of this ordering: on $\{p_{(1)}\}$, then $\{p_{(1)}, p_{(2)}\}$, $\{p_{(1)}, p_{(2)}, p_{(3)}\}$ and so on. For prefix length k the statistic is $S_k \sim \chi_k^2$ under the null, and each is converted to $-\log_{10}(p)$. The subset size at which this curve attains its maximum is denoted peak k , and the corresponding prefix is the subset of top-scoring phenotypes giving the strongest joint association for that SNP.

Computing each of these 1,010 nested statistics independently would require a fresh $O(k^3)$ Cholesky factorization at every prefix length. This is wasteful since the covariance submatrix for a prefix of length $k - 1$ is exactly the leading $(k - 1) \times (k - 1)$ block of the length- k submatrix, so almost all of the work done at step $k - 1$ can be reused at step k rather than discarded. Concretely, if L_{k-1} is the Cholesky factor of Σ_{k-1} , the Cholesky factor L_k of Σ_k has the block form:

$$L_k = \begin{pmatrix} L_{k-1} & 0 \\ \ell^\top & L_{kk} \end{pmatrix},$$

i.e. its leading $(k - 1) \times (k - 1)$ block is exactly L_{k-1} , unchanged from the previous step, so only the new row $[\ell^\top, L_{kk}]$ needs to be computed (full derivation in Appendix D). This row follows by solving $L_{k-1}\ell = \Sigma_{1:k-1,k}$ via forward substitution, at $O(k^2)$ cost rather than the $O(k^3)$ a fresh factorization would require, and setting $L_{kk} = \sqrt{\Sigma_{kk} - \ell^\top \ell}$.

The statistic updates the same way. Writing $Y_k = (Y_{k-1}^\top, \alpha_k)^\top$, where $Y_k = L_k^{-1} \mathbf{z}_{(k)}$ is the solution vector at prefix length k : the first $k - 1$ entries of Y_k are exactly Y_{k-1} , already computed at the previous step, and the new entry is $\alpha_k = (z_{(k)} - \ell^\top Y_{k-1})/L_{kk}$ which is the part of phenotype k 's z-score not already predicted by the phenotypes ahead of it in the ranking. Under the null each α_k is an independent standard normal, so $S_k = S_{k-1} + \alpha_k^2$ is exactly the running sum of squares required for $S_k \sim \chi_k^2$ to hold at every prefix length.

Net effect. Walking through the score-ranked ordering once, updating L by one row and S by one term at each step, produces the entire curve $S_1, \dots, S_{1010}$ for a SNP at a cost of $O(k^2)$ per step, rather than a fresh $O(k^3)$ factorization at each of the 1,010 prefix lengths individually. This is the saving that makes the cumulative test tractable across all 9,611,261 SNPs.

Early stopping. The cost of step k is still $O(k^2)$, so evaluating every curve to $k = 1,010$ is dominated by prefixes far beyond where the peak occurs. The loop therefore exits once the statistic falls more than 30% below the highest value reached so far, subject to a minimum of five phenotypes, and the peak is taken as that running maximum. Across the full dataset this reduces the work to approximately 3% of the full curve. The rule returns the global maximum whenever the curve is unimodal, but returns the first local maximum otherwise, so a curve that declines and later recovers will have its peak recorded at the earlier of the two. The same rule and tolerance are applied to the null run, so both compute the same statistic.

Ridge regularization. Positive definiteness of Σ_k is required at every step of the update above, so a ridge of 10^{-4} is added to the diagonal of $\hat{\Sigma}$ before factorization. The smallest eigenvalue of the full $1,010 \times 1,010$ matrix is 1.42×10^{-3} , an order of magnitude above the ridge, so it does not materially alter the covariance structure. This comparison is in fact the conservative, worst-case one: by Cauchy's interlacing theorem, the smallest eigenvalue of any $k \times k$ principal submatrix of a

symmetric matrix is always at least as large as that of the full matrix, so no score-ranked prefix of any size k encountered in the incremental construction above can have a smaller minimum eigenvalue than 1.42×10^{-3} . Benchmarking the ridge against the full matrix is therefore the hardest case the incremental pipeline could ever encounter, not merely a representative one. Tail probabilities are evaluated with `scipy.stats.chi2.logsf`, which falls back to a normal approximation when it underflows; this fallback was reached for 129 of the 9,611,261 real SNPs and for none of the null SNPs, and those 129 peaks were recomputed exactly using the regularised upper incomplete gamma function at 80 digits of precision. The corrections were substantial: the largest, for `rs10699448`, moved from an apparent 32614.47 to an exact 330.45.

Null calibration. The ranking above is derived from the phenotype score, which is itself computed from the same z-scores that then enter the joint test. The phenotypes are therefore selected on the data and the resulting p-values are not valid as stated. To measure the size of this effect, a null dataset of the same dimensions as the real one was simulated: 9,611,261 z-vectors drawn as $\mathbf{z} \sim \mathcal{N}(0, \hat{\Sigma}_{1010})$, carrying the phenotype correlation structure of the real data but no genetic signal. These were passed through the complete pipeline: the same 10,000 subsets as the production run, drawn from the same seed, followed by the same omnibus test, rescaling, phenotype score computation, and cumulative construction. Since the simulated data contains no association and no genetic effect built in by construction, the peak of each null curve measures inflation alone.

Each real SNP’s peak is then converted to an empirical p-value against this null distribution,

$$p_{\text{emp}} = \frac{1 + \#\{n : \text{peak}_n^{\text{null}} \geq \text{peak}^{\text{obs}}\}}{1 + N_{\text{null}}},$$

where the offsets prevent an empirical p-value of exactly zero. With $N_{\text{null}} = 9,611,261$ the smallest attainable value is 1.04×10^{-7} , so SNPs exceeding the largest null peak are reported as a count rather than ranked among themselves. Simulating the null at the same size as the real dataset rather than at a smaller scale is what brings this floor low enough to be useful genome-wide: the expected number of real SNPs exceeding the null maximum by chance is $N_{\text{SNP}}/(N_{\text{null}} + 1) \approx 1$.

2.5 Implementation and validation protocol

Since each subset requires only a single $O(k^3)$ Cholesky factorization regardless of the number of SNPs, the natural axis to parallelize over is subsets, not SNPs. Growing the SNP count increases the cost of each subset’s batched triangular solve linearly, but does not change how many independent units of work there are to distribute across workers.

At full scale the 10,000 subsets were divided into 100 chunks of 100, distributed across 64 worker threads via `joblib`. Holding the entire $k \times M$ arrays in memory simultaneously is not feasible at this scale: with $M \approx 9.6$ million SNPs and k up to 90, this would require tens of gigabytes per subset per worker thread. The sampling loop therefore processes SNPs in contiguous blocks of 2,000,000 columns at a time, capping the per-thread memory footprint regardless of the total SNP count. Since the Cholesky factor L depends only on the phenotype subset and not on the SNP values, it is computed once per subset outside the batch loop and reused across all five batches needed to cover the 9.6 million SNPs, giving exactly 10,000 factorizations and 50,000 batched triangular solves in total. Within each batch, the squared row norms are accumulated directly into a pre-allocated `float32` buffer via `np.einsum`, avoiding the intermediate array that a naive `(Y**2).sum(axis=0)` would materialize, and selected phenotype indices are sorted in ascending order beforehand to improve cache locality when gathering the corresponding covariance matrix rows.

Each worker writes its assigned rows of `stat_matrix.dat` directly to the shared memory-mapped file, so no synchronisation between threads is required and the full matrix is never held in RAM. The `sel_matrix` is accumulated in RAM and saved separately. Computing all $10,000 \times 9,611,261 \approx 9.6 \times 10^{10}$ test statistics completed in 5 hours 17 minutes on 64 CPU cores, with peak memory usage of approximately 100 GB. The phenotype score matrix was computed separately over 4 worker processes in batches of 500,000 SNPs, with both input and output read from and written to memory-mapped files, and completed in 9.3 minutes. This is markedly faster than the sampling stage itself, reflecting that the score computation is a single pass of dense matrix multiplication over already-computed statistics. Runtimes and hardware for the intermediate stages are given in Appendix D.

Every optimization described above was validated against a naive reference implementation before being used. The reference forms $\hat{\Sigma}_k^{-1}$ explicitly via `np.linalg.inv`, computes the statistic directly, and rescales using `scipy.stats`, deliberately using no Cholesky factorization and no `scipy.special` functions. Comparing the two methods across 100 independently drawn subsets, 50,000 SNPs and all 1,010 phenotypes, the selection matrices matched with the maximum absolute difference in the rescaled statistic being 7.52×10^{-4} (mean 1.24×10^{-6}), attributable solely to `float32` rounding between the two numerically distinct algorithms.

This protocol identified a second numerical issue in addition to the `float32` underflow described in Section 2.3.2. When `np.einsum` is called with an `out` argument on a reduction, it adds to the existing contents of the output array rather than replacing them. Since the output buffer is pre-allocated once per chunk and reused across all subsets within it, each subset was accumulating contributions from all previous subsets in the chunk. For later subsets the inflated values pushed survival probabilities below 10^{-300} , pinning the output to the clip floor for all but the weakest SNPs. The fix was to zero the buffer at the start of each subset’s batch loop. Both fixes were applied before the full-scale run.

After the run, `sel_matrix` was checked in full and contained no missing values and only the expected values $\{0, 1\}$. Since `stat_matrix.dat` occupies approximately 384 GB on disk, it was checked via a random sample of 500 subsets and 10,000 SNPs drawn with a fixed seed. The sampled block contained no missing values, no infinities, and no exact-zero entries, with values ranging from 0 to 840.01.

3 Results

3.1 Stability and convergence of the sampling procedure

Exhaustive enumeration is only tractable for very small phenotype pools. For $K = 5$ the total number of non-empty subsets is $\sum_{k=1}^5 \binom{5}{k} = 31$, and for $K = 10$ it is 1,023. At these scales every subset can be tested against every SNP, and the subset yielding the largest statistic for a given SNP is identified directly by a column-wise `argmax` over the full statistic matrix, with the corresponding phenotype membership read from `sel_matrix`. The optimal subset is therefore obtained by construction, since no subset is left untested.

This does not extend from $K = 100$ onwards, and the sampling procedure of Section 2.3 is used at every scale beyond the two pilot stages. The question that follows is how many subsets must be drawn for the resulting estimates to be stable.

To assess this, $N_{\text{rep}} = 10$ independent replicates of the production sampling procedure were run for a single SNP, each replicate using an independently seeded generator so that the replicates are genuine re-runs of the sampling procedure rather than resamples of a shared pool. Within a replicate, subsets are drawn one at a time and a running per-phenotype average score is maintained, checkpointed at $N_s \in \{1,000, 5,000, 7,500, 9,000, 10,000\}$ subsets. This yields, for every phenotype and every checkpoint, 10 independent estimates of the phenotype score at that sample size, whose spread measures how much the estimate would have varied had a different random draw of subsets been used. SNP `rs80170726` was randomly selected for this inspection, covering all 1,010 phenotypes as well as a restricted view of the 10 phenotypes with the highest score at $N_s = 10,000$.

Figure 3 shows the 10 replicate score estimates at each checkpoint for the top-10 scoring phenotypes. Across all of them, the spread of the boxes shrinks substantially from $N_s = 1,000$ to $N_s = 10,000$, and the medians stabilise well before the full 10,000 subsets are reached. The same pattern generally holds across the full 1,010-phenotype view. The phenotype score is therefore stable at $N_s = 10,000$, despite this being a vanishingly small fraction of the $\approx 10^{304}$ possible subsets, because the score is an average over the subsets containing a given phenotype and averages converge quickly.

This check used only $N_{\text{rep}} = 10$ replicates, and it recomputes the sampling procedure independently rather than reusing the stored `stat_matrix` and `sel_matrix` from the production run, so it verifies the convergence behaviour of the method rather than the specific values held on disk. It supports, but does not by itself prove, that $N_s = 10,000$ is an adequate sample size at every SNP and moreover it becomes an open question to whether we could set $N_s < 10,000$; there is a similar level of variability for $N_s = 7,500$ or $N_s = 9,000$ across the phenotypes. It also concerns the phenotype score specifically. The minimum subset p-value used in Section 3.2 is based on the maximum

statistic across sampled subsets rather than an average, and an extreme value does not carry the same convergence guarantee.

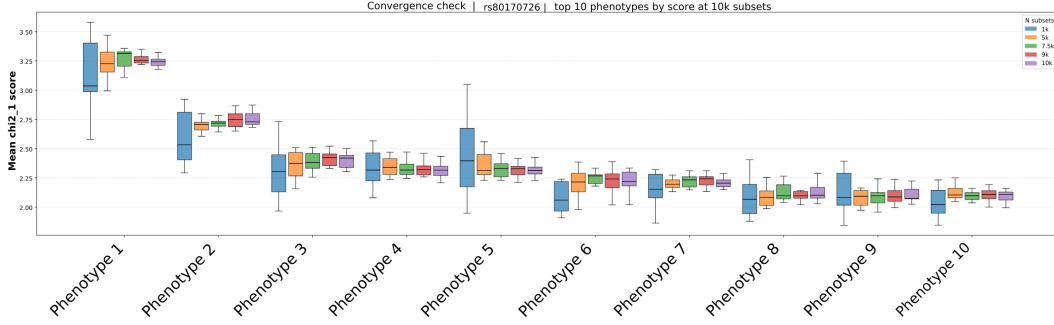


Figure 3: Convergence of the per-phenotype score estimate for SNP rs80170726, top 10 phenotypes by score at $N_s = 10,000$. Each box summarizes 10 independent replicates of the sampling procedure, checkpointed at increasing numbers of subsets N_s .

3.2 Comparing univariate, joint, and subset-based p-values

With both the z-score matrix and the full-scale sampling output available, three different per-SNP significance tests can be compared against one another: the standard single-phenotype univariate test, a joint test using all 1,010 phenotypes simultaneously, and a Bonferroni-corrected test based on the best of the 10,000 random subsets. The aim is to see whether the random subset sampling strategy finds association signal that the other two, more standard, approaches miss.

For each SNP j , the three p-values are computed as follows. The **minimum univariate p-value** tests each phenotype k independently, with $(z_{j,k})^2 \sim \chi_1^2$ under the null, and reports $\min_k p_{j,k}$, the strongest single-phenotype association for SNP j . This ignores correlations between phenotypes entirely. The **joint p-value** decorrelates the full 1,010-length z-score vector using the Cholesky factor L of the LDSC intercept matrix, giving $Y_j = L^{-1}z_j$ and $S_j = \|Y_j\|^2 \sim \chi_{1010}^2$ under the null. The **minimum subset p-value** takes the maximum of the 10,000 rescaled statistics already stored in `stat_matrix`, converts it back to a p-value, and multiplies it by 10,000, clipped at 1, to correct for having tested 10,000 subsets per SNP [4]. This answers whether any sampled subset gives a significant result for SNP j , once the multiple-testing penalty for trying 10,000 of them is applied.

All three columns were computed in a single combined pass over 4 worker processes, each handling a contiguous quarter of the 9,611,261 SNPs in batches of 200,000. Both the z-score matrix and `stat_matrix` are read within the same batch loop rather than in two separate scans, halving the I/O relative to computing the univariate, joint and subset-based columns independently. The run completed in around 124 minutes.

Table 1 breaks the full set of 9,611,261 SNPs down by every combination of significance ($p < 5 \times 10^{-8}$) across the three tests.

Category	SNPs (% of total)
None significant	7,666,679 (79.768%)
Univariate only	4,391 (0.046%)
Joint only	1,779,787 (18.518%)
Subsets only	259 (0.003%)
Univariate + Joint	12,805 (0.133%)
Univariate + Subsets	407 (0.004%)
Joint + Subsets	82,215 (0.855%)
All three	64,718 (0.673%)

Table 1: Contingency table of significance across the three tests, out of all SNPs.

The joint test alone accounts for the largest block of unique detections (18.518%), consistent with it being the most powerful of the three when a SNP affects many phenotypes concordantly. The subset test's unique contribution is small by comparison: 259 SNPs reach significance only through the subset test, and a further 407 are also univariate-significant but still missed by the joint test. Together these 666 SNPs (0.45% of the 147,599 significant under the min-subset-Bonferroni test) are subset-specific i.e., significant under the subset test but not the joint test, irrespective of univariate status. This identifies SNPs for which some particular combination of phenotypes carries a signal that is diluted away once all 1,010 phenotypes are tested jointly.

Figure 4 plots, for each SNP, the joint and minimum-subset $-\log_{10}$ p-values against the corresponding univariate $-\log_{10}$ p-value, with the $y = x$ line as the reference against which points are read as more or less significant. Points above the diagonal indicate a more significant result than the best single phenotype. Most points fall above the diagonal for the joint test, indicating that decorrelating and combining all 1,010 phenotypes typically yields a more significant result than the best single phenotype alone. The minimum-subset test falls closer on average to the diagonal, consistent with it being a noisier, sampling-based approximation of the same idea restricted to 10,000 random combinations rather than the single full combination. This suggests that there exist subset combinations carrying stronger signal that are not covered by the current 10,000 random subsets, and/or that increasing N_s and the subset size range is worth pursuing.

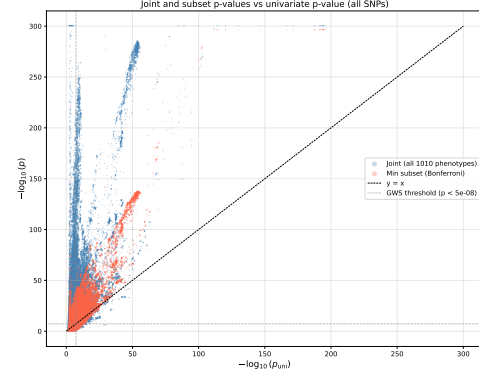


Figure 4

The minimum-subset test is Bonferroni-corrected for 10,000 subsets, which is conservative given that the subsets are highly overlapping in phenotype membership rather than independent tests. The true effective number of tests is smaller, so this correction likely understates the significance of the minimum-subset p-value. This comparison also does not identify which phenotypes drive a SNP's subset-based signal, or which SNPs are significant only under the subset-based test and not the other two, both of which are addressed in the following subsections.

3.3 Phenotype score vs. univariate signal

The comparison in Section 3.2 operates at the level of whole SNPs. This subsection asks a finer question: whether the phenotype score captures association signal beyond what univariate testing detects, and in particular whether any phenotypes score highly in the subset test despite weak marginal z-scores.

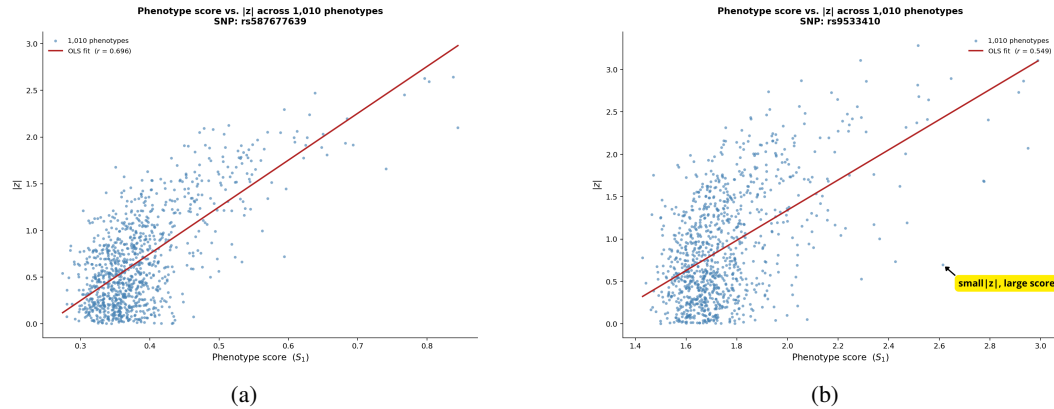


Figure 5: Phenotype score vs. $|z|$ across all 1,010 phenotypes for two representative SNPs, with OLS trend line in red. (a) rs587677639 ($r = 0.696$): score and $|z|$ track closely, as expected. (b) A discordant SNP ($r = 0.549$): phenotypes in the lower right score highly despite small $|z|$, signal invisible to univariate testing.

For a given SNP, each of the 1,010 phenotypes can be placed in a scatter plot by its phenotype score at that SNP against its marginal $|z|$ at that SNP. Figure 5a shows a representative SNP, rs587677639, where the two quantities have a moderately high positive correlation ($r = 0.696$): phenotypes that score highly in the subset test also tend to have a strong univariate signal at that SNP. This is the expected pattern, since a phenotype with a large $|z|$ contributes a large term to the joint statistic of every subset it is drawn into, and the phenotype score averages exactly those subsets.

Figure 5b shows a different SNP where this relationship breaks down ($r = 0.549$). A cluster of phenotypes in the lower right of the plot has a large phenotype score but a small $|z|$: their association with this SNP is not detectable in a single-trait test, yet the subset test consistently assigns them an elevated score. These phenotypes are found by subset testing precisely because their signal concentrates only when tested jointly with the right combination of other phenotypes. This is something neither a univariate test nor a joint test over all 1,010 phenotypes would reveal.

3.4 Cumulative joint testing of score-ranked phenotypes

The omnibus test does not identify which phenotypes within a subset carry a SNP’s signal (Section 2.3.1). Section 2.4 constructs the subset directly by ranking the 1,010 phenotypes by their phenotype score and computing the joint statistic cumulatively, via the incremental Cholesky update, over nested prefixes of that ranking, with the subset size at the peak of the resulting curve denoted peak k . This is not evaluated to $k = 1,010$ for every SNP: as described in Section 2.4, the loop exits once the curve falls 30% below the highest value reached so far, which reduces the cost per SNP by roughly two orders of magnitude and makes the analysis tractable across the full dataset.

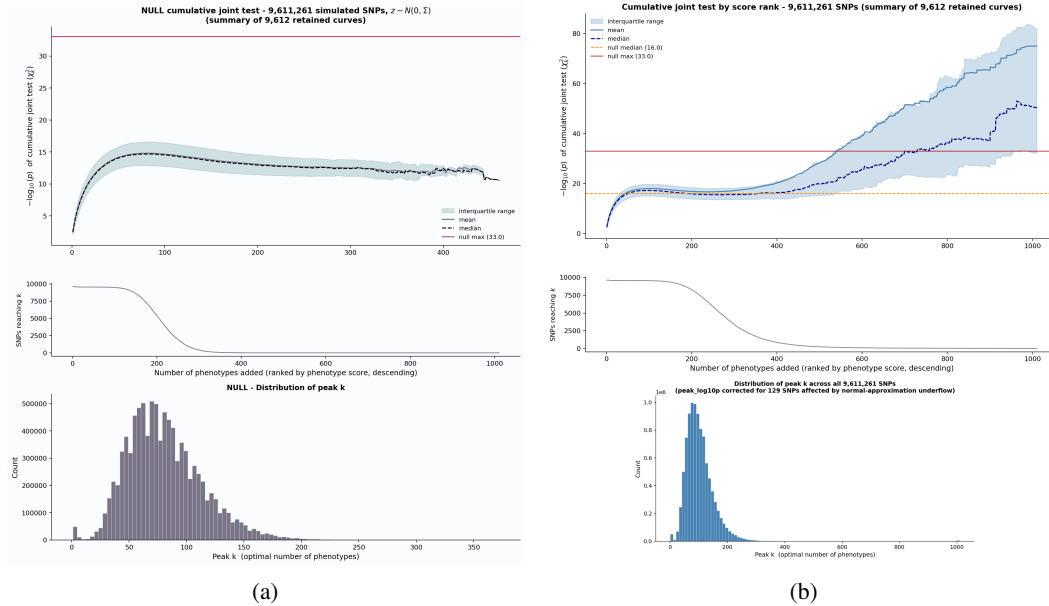


Figure 6: Cumulative joint test of score-ranked phenotypes, for (a) 9,611,261 simulated null SNPs drawn as $\mathbf{z} \sim \mathcal{N}(0, \hat{\Sigma}_{1010})$ and (b) 9,611,261 real SNPs. Top: $-\log_{10}(p)$ against the number of top-scoring phenotypes included, summarized as mean and median across the retained curves. Middle: number of SNPs whose curve was still running at each k , since early stopping ends curves at different points. Bottom: distribution of peak k .

Since the ranking is derived from the same z -scores that then enter the joint test, the phenotypes are selected on the data and the resulting p -values are not valid as stated. Ranking phenotypes by score puts those with the largest z -scores first, so the top-ranked subsets are built from whatever is most extreme in that SNP’s z -vector, and the joint statistic then rediscovers exactly what the ranking used to order the phenotypes. This inflates the p -value by an amount that has nothing to do with association, and that amount must be measured before any peak can be interpreted.

The null calibration described in Section 2.4 measures it directly. 9,611,261 null $\mathbf{z}$ -vectors were drawn as $\mathbf{z} \sim \mathcal{N}(0, \hat{\Sigma}_{1010})$, matching the real dataset in size and carrying the same phenotype correlation structure but no genetic signal, and passed through the entire pipeline (Figure 1) with the same subsets, the same omnibus test and rescaling, the same phenotype score computation, and the same stopping rule. Any peak the procedure produces from these is inflation and nothing else. Matching the null to the real dataset in size is what makes the comparison useful at genome-wide scale: with $N_{\text{null}} = 9,611,261$ the empirical p-value of Equation (2.4) resolves to 1.04×10^{-7} , so approximately one real SNP would be expected to exceed the null maximum by chance.

Figure 6 shows the two runs side by side, and the difference is in the shape of the curves rather than their height at the peak. Both rise steeply over the first few dozen phenotypes and turn over near $k \approx 80$, and up to $k \approx 300$ the real curves sit only slightly above the null. Beyond that the two diverge completely. The null curves decline monotonically and every one of the 9,611,261 null SNPs triggered the stopping rule, the last at $k = 552$; no null curve ever recovers once it has turned over. The real curves instead climb again from $k \approx 400$ onwards, and the mean crosses the null maximum of 33.05 at around $k = 530$ and continues to roughly 75 by $k = 1,010$. This upturn is the signature of genuine multitrait signal: phenotypes ranked far down the list still contribute association rather than only degrees of freedom, which cannot happen when the $\mathbf{z}$ -vector is pure noise. It is also why 0.4% of real SNPs never trigger the stopping rule and run the full length of the phenotype pool.

Under the null the peak reaches a median $-\log_{10}(p)$ of 16.03, and across all 9,611,261 simulated SNPs it never exceeds 33.05 (Table 2). Ranking phenotypes on the data and then testing them therefore produces p-values around 10^{-16} from pure noise, and 33.05 is the reference against which the real peaks are judged.

Table 2: Peak $-\log_{10}(p)$ of the cumulative joint test across 9,611,261 real SNPs and 9,611,261 simulated null SNPs.

	median	p95	p99	max	peak k (median)
Null (simulated)	16.03	20.84	23.05	33.05	76
Real SNPs	18.80	27.08	38.22	1881.25*	94

*This peak occurs at $k = 1007$, essentially the full joint test, and reflects amplification through a low-eigenvalue direction of $\hat{\Sigma}^{-1}$ rather than a uniquely strong association: no phenotype for this SNP reaches genome-wide significance ($\max |z| = 3.35$).

Table 2 quantifies what the figure shows. The bulk of the real distribution falls inside the null range: 95% of real SNPs peak below 27.08 against a null maximum of 33.05, so most of what the cumulative test appears to find in the real data is attributable to selection rather than association. The tail is different. The real 99th percentile of 38.22 already exceeds the null maximum, and the distribution continues well beyond it, reaching 90.62 at the 99.9th percentile against a null 99.99th percentile of 27.98. Converting each real peak to an empirical p-value via Equation (2.4), 173,033 SNPs (1.80%) exceed the maximum observed across all 9,611,261 null draws, where approximately one would be expected by chance. The selection procedure alone never produced a peak above 33.05 in 9,611,261 attempts, so these SNPs cannot be attributed to it.

Eigenvalue contamination check. A near-singular $\hat{\Sigma}_k$ can inflate the statistic $S = \mathbf{z}_k^\top \hat{\Sigma}_k^{-1} \mathbf{z}_k$ through a direction of near-zero eigenvalue rather than through genuine coordinated signal, so this was checked directly rather than assumed. For each of the 173,033 hits, the peak-subset covariance matrix was eigendecomposed and the fraction of S attributable to directions with $\lambda < 10^{-2}$ was computed. This fraction is concentrated near zero across the discovery set (median 0.000, 90th percentile 0.011, 99th percentile 0.062), so eigenvalue degeneracy is not a broad driver of the 173,033 hits. A small minority, 26 SNPs (0.015%), have more than half their statistic explained this way and are flagged as artifacts rather than genuine coordinated effects; excluding them leaves a clean discovery count of 173,007 (1.80%).

Interpretation. Not all 173,007 clean discoveries are subset findings in the sense motivating this analysis. 10,073 (5.8%) peak at $k > 950$, where the cumulative test is effectively the joint test over

all 1,010 phenotypes and their significance is already captured by the joint column of Section 3.2. Direct inspection of two representative cases in this region, rs3217089 and rs76016924, illustrates the exact mechanism the check above screens for. Neither has any phenotype reaching genome-wide significance ($\max |z| = 3.35$ and 3.49), and their summed squared z-scores are only $2.19\times$ and $1.10\times$ the null expectation, yet their joint statistics are inflated $12.1\times$ and $3.5\times$.

The source is the eigenstructure of $\hat{\Sigma}_k$. Writing its orthonormal eigenvectors as $\mathbf{v}_i$ with eigenvalues λ_i , each $\mathbf{v}_i$ is a weighted combination of the k phenotypes and λ_i is the variance of the z-scores along it, so a small λ_i marks a combination in which the phenotypes are nearly redundant. Since the inverse is diagonal in this basis, the statistic separates into independent contributions,

$$S = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z} = \sum_{i=1}^k \frac{(\mathbf{v}_i^\top \mathbf{z})^2}{\lambda_i},$$

each with expectation 1 under the null. Dividing by λ_i is the intended behaviour, since signal along a usually-redundant direction is genuinely surprising, but it leaves the statistic sensitive to how accurately the smallest eigenvalues have been estimated. For rs3217089 the single smallest-eigenvalue direction contributes 44.4% of the statistic and the 25 directions with $\lambda < 10^{-2}$ contribute 63.8% and for rs76016924; 33.9% and 45.5%, respectively (`inspect_large_k_snps.py`). The projections themselves are unremarkable, $(\mathbf{v}_i^\top \mathbf{z})^2 = 8.89$ and 3.27 for the two dominant directions, entirely consistent with the χ_1^2 they follow under the null. It is division by a near-zero eigenvalue, not an unusual projection, that produces the extreme statistic. This is also why the diagonal ridge of 10^{-4} added to $\hat{\Sigma}$ before factorization (Section 2.4) is not merely a numerical convenience: by placing a floor under how small λ_i can become, it directly bounds how much any single eigendirection can inflate S , at the cost of a small, quantified bias in the null covariance, and the scan above confirms that bound holds in practice, since only 26 of 173,033 hits still breach it.

Limitations. The empirical p-value of Equation (2.4) cannot resolve below 1.04×10^{-7} , so all 173,033 SNPs above the null maximum receive the same value regardless of how far above it they fall and are reported as a count rather than ranked. This floor is above the conventional genome-wide threshold of 5×10^{-8} , so per-SNP genome-wide significance cannot be claimed from this calibration; the expected false-positive count of approximately one across the whole dataset is the appropriate statement instead. The simulated null SNPs are drawn independently, whereas the real SNPs include variants in LD, so the real dataset does not consist of 9,611,261 independent observations, and converting these counts into an estimate of independent loci (e.g. via distance- or LD-based clumping) is left to future work. Finally, 37,756 SNP identifiers are shared by more than one row, consistent with multi-allelic sites where several alternate alleles are assigned the same rsID; each row is treated as an independent variant throughout, and variants are referenced by row position rather than by identifier alone.

3.5 Comparing univariate, joint, and optimal-subset p-values

Section 3.2 compared univariate, joint, and subset-based significance using the best of 10,000 randomly sampled subsets. This subsection repeats that comparison with the subset column replaced by the optimal subset of Section 3.4: the score-ranked prefix at which each SNP’s cumulative curve peaks. The three columns, together with one phenotype-score column per trait, n_{optimal} (subset size at the peak), and the peak statistic itself, were assembled into a single table across all 9,611,261 SNPs.

Because the ranking used to build the optimal subset is derived from the same z-scores that then enter the test, its nominal χ^2 tail is inflated by selection, exactly as in Section 3.4: applying the standard $p < 5 \times 10^{-8}$ threshold directly would flag 99.48% of all SNPs. Optimal-subset significance is therefore defined, as before, by exceeding 33.05, the largest peak produced by any of the 9,611,261 null draws under the identical selection procedure. Univariate and joint retain the standard genome-wide threshold, unchanged from Section 3.2.

Category	SNPs (% of total)
None significant	7,666,428 (79.765%)
Univariate only	4,776 (0.050%)
Joint only	1,747,185 (18.179%)
Optimal subset only	510 (0.005%)
Univariate + Joint	19,839 (0.206%)
Univariate + Optimal subset	22 (0.0002%)
Joint + Optimal subset	114,817 (1.195%)
All three	57,684 (0.600%)

Table 3: Contingency table of significance across the three tests (univariate, joint, calibrated optimal subset), out of 9,611,261 SNPs.

Of the 173,033 SNPs clearing the calibrated optimal-subset threshold, 172,501 (99.69%) are also joint-significant, so the optimal subset mostly sharpens a signal the joint test already detects rather than finding new territory. This is expected because the calibrated threshold is considerably stricter than the joint test’s fixed nominal cutoff. The remaining 532 SNPs (0.31%) are significant only under the optimal-subset test; three coincide with the eigenvalue-degenerate artifacts of Section 3.4, leaving 529 genuine subset-specific discoveries. This is directly comparable to the 666 subset-specific SNPs found in Section 3.2 from the randomly-sampled subset test. The two constructions: uniform random sampling over $k \in \{5, \dots, 90\}$ and constructive score-ranking up to $k = 1,010$, are built completely differently and draw from largely non-overlapping regions of the combinatorial space, yet arrive at a comparable order of magnitude of joint-invisible discoveries, evidence that subset-based testing surfaces a real, if narrow, category of signal rather than an artifact of either construction. The two groups also differ consistently with the subset-size argument of Section 3.4: subset-specific SNPs peak at a median $k = 94$, against $k = 133$ for SNPs where the joint test independently agrees – significant under the optimal subset and the independent joint test. Therefore, signal unique to subset testing concentrates in a smaller prefix than the full 1,010-phenotype join dilutes away.

Selection artifacts. Since the optimal-subset curve is optimized over prefixes that, when unimodal, extend to the full joint statistic at $k = 1,010$, its peak exceeds the joint statistic for the large majority of SNPs; for 40,399 SNPs (0.42%) the reverse holds, measured as the gap $g = \text{neglog10}p_{\text{joint}} - \text{neglog10}p_{\text{optimal}}$ turning positive. This is not misidentification of the optimal subset: n_{optimal} records only the k at which the 30%-drop early-stopping rule of Section 3.4 fired, not the true global maximum. For 78% of these SNPs ($n_{\text{optimal}} < 10$, median 3), the curve peaks within the first few score-ranked phenotypes, declines, and the search stops without ever reaching $k = 1,010$; the joint test, evaluated independently over the full phenotype set, then finds stronger signal the truncated search never had the chance to look for (median $g = 2.10$, up to $1,085 - \log_{10}(p)$ units). A separate group of 1,360 SNPs (3.4% of the 40,399) has $n_{\text{optimal}} = 1,010$ exactly meaning their curves never triggered early stopping (median $g = 0.86$). This is the signature of the 10^{-4} ridge added to $\hat{\Sigma}$ throughout the incremental Cholesky pipeline (Section 3.4), which the joint test’s unregularized $\hat{\Sigma}$ does not carry.

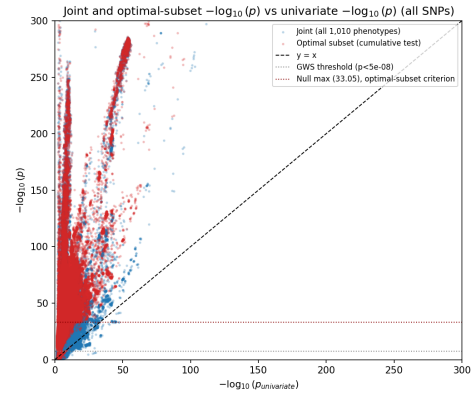


Figure 7: Joint and optimal-subset $-\log_{10}(p)$ per SNP against minimum univariate $-\log_{10}(p)$. Axes clipped to $[0, 300]$. Dotted lines mark the calibrated optimal-subset threshold (33.05) and the standard GWS threshold (5×10^{-8} , univariate/joint).

Figure 7 shows both series sitting almost entirely above the $y = x$ diagonal, confirming that combining phenotypes, whether via the full joint test or the constructed optimal subset, is essentially always at least as powerful as the best single phenotype, and that the optimal-subset series tracks slightly above the joint series at a given univariate value across most of the range.

4 Discussion

What the method provides. Taken purely on raw SNP count, subset-based testing is not the more powerful test: the joint test alone accounts for the largest block of unique detections (18.5% of all SNPs, Table 1), against 666 SNPs found uniquely by random subset sampling (Section 3.2) and 529 by the constructed optimal subset (Section 3.5). Where subset-based testing does contribute at the level of individual SNPs, it is narrow, and framing the method as a way to out-detect the joint test is an overstatement of what Sections 3.2 and 3.5 actually show.

Random subset sampling instead is answering a question neither univariate nor full-joint testing can: *which* phenotypes carry a SNP’s signal. The phenotype score (Section 2.3.3) surfaces traits that score highly in subset testing despite an at most ordinary marginal z-score (Section 3.3), something averaging over all 1,010 phenotypes at once necessarily erases. The cumulative construction of Section 3.4 sharpens this further: the constructed subset typically matches or exceeds full joint-test power using a median of only 94 of the 1,010 phenotypes, cross-validated independently by two very different subset constructions: random sampling and score-ranking, which arrive at a comparable order of magnitude of joint-invisible discoveries despite drawing from largely non-overlapping regions of the combinatorial space (Section 3.5). The practical value is therefore not that subset testing beats the joint test on raw count, but that it recovers essentially the same power from a small, named, and far cheaper subset of phenotypes. The difference between "some combination of the 1,010 phenotypes is associated with this SNP" and a short, concrete list of candidate drivers is something to follow up on.

That list should still be read as suggestive rather than confirmatory. Attribution here is descriptive: the phenotype score ranks phenotypes by their average contribution across sampled subsets, but this ranking does not need to agree with the specific phenotypes making up any one significant subset, and no single phenotype needs to dominate a SNP’s subset-specific signal the way one might hope. Identifying which phenotype, or combination of phenotypes, is mechanistically responsible for a SNP’s coordinated effect is not something either the omnibus test or the phenotype score can establish on its own.

Limitations. $N_s = 10,000$ is a vanishing fraction of the $\approx 10^{304}$ possible subsets, and Section 3.1’s convergence check applies to the phenotype score’s average, not to the extreme-value statistics built from it: the minimum-subset p-value of Section 3.2 takes a maximum over sampled subsets rather than an average, and a larger N_s could still reveal a stronger subset in the tail that 10,000 draws happened to miss. Subset size and membership are drawn uniformly at random throughout; an adaptive sampling scheme was not explored, and even the score-ranked construction of Section 3.4 is only informed along one axis (the data itself), not by any prior structure such as phenotype anatomy or LD. The minimum-subset test’s Bonferroni correction for 10,000 subsets is conservative, since the subsets are highly overlapping rather than independent, which likely understates the significance it reports.

The null covariance estimator relies on every phenotype sharing the same 32,947 individuals (Section 2.2), which is what lets the LDSC sample-overlap term collapse to the phenotypic correlation matrix. However, this simplification does not hold for meta-analyzed or partially-overlapping cohorts, where the full sample-overlap term would need to be estimated rather than assumed away. Of the 9,611,405 candidate SNPs, 144 with any missing phenotype value were dropped outright rather than handled per-subset, and 37,756 SNP identifiers are shared across more than one row (multi-allelic sites), addressed here only by treating each row as an independent variant. Finally, unlike the simulated nulls, the real dataset’s 9,611,261 SNPs are not independent observations as they are correlated through LD so the discovery counts reported throughout Section 3 describe correlated SNPs rather than independent genetic signals; converting them into an estimate of independent loci via distance- or LD-based clumping is left to future work.

Numerical reproducibility. Two bugs survived what looked, on inspection, like a numerically straightforward implementation. The first was `float32` underflow in the survival function used for rescaling (Section 2.3.2): tail probabilities below 10^{-38} underflow to exactly 0.0 in `float32`, silently defeating the intended clip and propagating `inf`. The second, found only by comparing against a deliberately naive reference implementation (Section 2.3), was that `np.einsum` called with an out argument on a reduction accumulates into the existing buffer rather than overwriting it; with that

buffer reused across every subset in a chunk, later subsets inherited the accumulated contributions of every subset before them. This second bug is the more instructive one: it produced plausible-looking wrong values, pinned near a numerical clip boundary rather than obviously corrupted, and would not have been caught by inspection alone. Both were only found by validating each optimization against a slower, transparent reference before trusting it at the next scale. The same protocol is the reason to trust the results reported here, and is worth treating as a required step rather than an optional check whenever an implementation is optimized for scale.

Outlook. A wider and adaptive subset-size search, informed by the peak- k distribution already in hand (Table 2), could test whether the 259 subset-only SNPs of Section 3.2 grow with a larger N_s and size ceiling, or whether the 173,033-SNP count of Section 3.4 shifts. Converting these discovery counts into independent loci via LD- or distance-based clumping, and extending the null covariance estimator beyond the single-cohort case to meta-analyzed or partially-overlapping studies, are both natural next steps that the current single-cohort, single-ancestry scope does not address. Packaging the pipeline itself into a documented, maintained tool rather than a sequence of per-stage scripts is worth pursuing separately, but is a software-engineering task rather than a scientific one.

5 Conclusion

This article implements, validates, and scales a random subset sampling strategy for multitrait GWAS to the full UK Biobank brain MRI dataset, and shows that constructing a SNP’s phenotype subset directly, by ranking phenotypes by score and testing cumulatively, recovers essentially the same power as the full 1,010-phenotype joint test from a small, interpretable fraction of it. The method’s value is not that it finds more significant SNPs than the joint test, but that it narrows down which phenotypes are actually driving a SNP’s signal, giving a short list worth following up on instead of one out of all 1,010 phenotypes.

One further improvement is worth noting beyond those raised in Section 4. Rather than dropping the 144 SNPs with any missing phenotype value (Section 2.1), a per-subset filtering pipeline instead keeps a SNP for a given subset only if every phenotype in that subset is observed for it, excluding a SNP only from the specific subsets it cannot support, at the cost of subset counts that scale with the matrix’s sparsity (an estimated $1.1\text{--}1.5\times$ the baseline N_s). This pipeline was implemented and tested on an independent dataset of 78 GWAS of quantitative traits and diseases from the UK Biobank, to confirm the method generalizes beyond the single, fully-observed brain MRI phenotype pool used throughout this article, and reproduced the full suite of analyses and figures presented here. All scripts, covering both sparse and fully dense (0% missing) input z-score matrices, are available on GitHub.¹

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A LDSC intercept as an unbiased estimator of the null covariance

Proof. For K phenotypes and M SNPs, the marginal GWAS estimator at SNP j for phenotype k is $\hat{\beta}_{j,k} = \mathbf{x}_j^\top \mathbf{y}_k / N_k$ under the linear model $\mathbf{y}_k = \mathbf{X}\boldsymbol{\beta}_k + \boldsymbol{\varepsilon}_k$, $\boldsymbol{\varepsilon}_k \sim \mathcal{N}(0, \sigma_k^2 \mathbf{I})$. Under the polygenic assumption that any single SNP explains a negligible fraction of phenotypic variance, $\sigma_k^2 \approx 1$, and the z-score reduces to $z_{j,k} \approx \sqrt{N_k} \hat{\beta}_{j,k}$.

The (i, k) entry of the null covariance matrix is $\mathbb{E}[z_{j,i} z_{j,k}]$. Substituting the linear model into both z-scores and expanding the product gives

$$z_{j,i} z_{j,k} = \frac{1}{\sqrt{N_i N_k}} \left[\underbrace{(\mathbf{x}_j^\top \mathbf{X} \boldsymbol{\beta}_i)(\mathbf{x}_j^\top \mathbf{X} \boldsymbol{\beta}_k)}_{\text{genetic}} + \underbrace{(\mathbf{x}_j^\top \mathbf{X} \boldsymbol{\beta}_i)(\mathbf{x}_j^\top \boldsymbol{\varepsilon}_k) + (\mathbf{x}_j^\top \boldsymbol{\varepsilon}_i)(\mathbf{x}_j^\top \mathbf{X} \boldsymbol{\beta}_k)}_{\rightarrow 0} + \underbrace{(\mathbf{x}_j^\top \boldsymbol{\varepsilon}_i)(\mathbf{x}_j^\top \boldsymbol{\varepsilon}_k)}_{\text{intercept}} \right].$$

The two cross terms vanish in expectation since effect sizes and residuals are independent with $\mathbb{E}[\boldsymbol{\varepsilon}] = 0$; what remains are a genetic term and a noise (intercept) term, evaluated separately below.

Intercept term. Since $\mathbf{x}_j$ is standardized and independent of the residuals, $\mathbb{E}[(\mathbf{x}_j^\top \boldsymbol{\varepsilon}_i)(\mathbf{x}_j^\top \boldsymbol{\varepsilon}_k)] = \mathbb{E}[\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_k]$, which is non-zero only for individuals measured in both studies i and k : for each of the $n_{i,k}$ shared individuals, the residuals covary at the phenotypic correlation $\rho_{i,k}$, giving $\mathbb{E}[\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_k] = N_{\text{overlap}}^{i,k} \rho_{i,k}$ and

$$[\Sigma_{\text{int}}]_{i,k} = \frac{N_{\text{overlap}}^{i,k} \rho_{i,k}}{\sqrt{N_i N_k}}.$$

This term depends only on sample overlap and phenotypic correlation, not genetics.

Genetic term. Under the infinitesimal polygenic model [1], effect sizes are independent across SNPs with $\text{Cov}(\beta_{l,i}, \beta_{m,k}) = (\sigma_{i,k}/M) \mathbf{1}_{l=m}$. Substituting this into the genetic term’s double sum collapses it to a single sum over l , since every cross-SNP term vanishes by independence, leaving $\sqrt{N_i N_k} (\sigma_{i,k}/M) \sum_l r_{j,l}^2$. That remaining sum is, by definition, the LD score $\ell_j = \sum_{l \neq j} r_{j,l}^2$ (the $l = j$ self-term contributes a negligible constant), so the genetic term is exactly linear in ℓ_j :

$$\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i} \beta_{m,k}] = [\Sigma_g]_{i,k} \ell_j, \quad [\Sigma_g]_{i,k} = \frac{\sqrt{N_i N_k} \sigma_{i,k}}{M}.$$

Combining both terms gives the LDSC decomposition used throughout Section 2.2:

$$\mathbb{E}[z_{j,i} z_{j,k}] = [\Sigma_g]_{i,k} \ell_j + [\Sigma_{\text{int}}]_{i,k}.$$

This says $z_{j,i} z_{j,k}$ is, across SNPs, exactly linear in ℓ_j with a mean-zero, LD-score-independent residual – precisely the model LDSC regression fits. The true parameters therefore lie within the fitted model family, so by standard OLS unbiasedness under a well-specified linear model, the OLS intercept $\hat{c}_{i,k}$ from regressing $z_{j,i} z_{j,k}$ on ℓ_j across all M SNPs satisfies $\mathbb{E}[\hat{c}_{i,k}] = [\Sigma_{\text{int}}]_{i,k}$, regardless of the genetic architecture $\sigma_{i,k}$: all LD-mediated genetic signal is absorbed into the slope, not the intercept. Assembled across all phenotype pairs, $[\hat{\Sigma}_{\text{int}}] = [\hat{c}_{i,k}]_{i,k}$ is therefore an unbiased estimator of the null covariance matrix $\hat{\Sigma}_K$ used in $S = \mathbf{z}^\top \hat{\Sigma}_K^{-1} \mathbf{z}$.

Single-cohort specialization. When all K phenotypes share the same cohort, $N_i = N_k = N_{\text{overlap}}^{i,k} = N$, the intercept collapses to $[\Sigma_{\text{int}}]_{i,k} = N \rho_{i,k} / \sqrt{N \cdot N} = \rho_{i,k}$: the null covariance matrix is exactly the $K \times K$ phenotypic correlation matrix, the reduction used in Section 2.2. $\square$

B Tail-preservation of the rescaling transform

Rescaling relies on two `scipy.special` functions: `chdtrc`(k, x), the χ_k^2 survival function, and `chdtri`(k, p), its inverse.

Proof. For $X \sim \chi_k^2$, integrating the density from x to ∞ and substituting $u = t/2$ gives

$$\text{chdtrc}(k, x) = P(\chi_k^2 > x) = Q\left(\frac{k}{2}, \frac{x}{2}\right),$$

the regularized upper incomplete gamma function. `chdtri`(k, p) is its inverse in x : the value x^* solving $Q(k/2, x^*/2) = p$, i.e. the inverse survival function $F_{\chi_k^2}^{-1}(1 - p)$.

Given the omnibus statistic $S_k \sim \chi_k^2$, rescaling applies the chain

$$S_k \xrightarrow{\text{chdtrc}(k, \cdot)} p = P(\chi_k^2 > S_k) \xrightarrow{\text{chdtri}(1, \cdot)} S_1 = F_{\chi_1^2}^{-1}(1 - p).$$

Since `chdtri`(1, $\cdot$) is exactly the inverse of `chdtrc`(1, $\cdot$), composing the two returns p unchanged: $P(\chi_1^2 > S_1) = p = P(\chi_k^2 > S_k)$. The transform therefore preserves the tail probability of S_k exactly, for any k , rather than approximating it – S_1 is simply S_k 's tail probability re-expressed on the χ_1^2 scale. $\square$

C Omnibus statistic via Cholesky factorisation

This appendix derives the result used throughout Section 2.4: Cholesky factorization reduces the omnibus statistic to a sum of squares, and this factorization can be updated incrementally, one phenotype at a time, without repeating the full $O(k^3)$ factorization at every prefix length.

Proof. Base case. For a fixed subset of k phenotypes, factorize $\Sigma_k = LL^\top$ with L lower triangular, and solve $LY = \mathbf{z}$ for $Y = L^{-1}\mathbf{z}$ by forward substitution at $O(k^2)$ cost rather than forming Σ_k^{-1} explicitly at $O(k^3)$. Since $\Sigma_k = LL^\top$, under H_0 ($\mathbf{z} \sim \mathcal{N}(0, \Sigma_k)$),

$$Y \sim \mathcal{N}(0, L^{-1}\Sigma_k(L^{-1})^\top) = \mathcal{N}(0, I_k),$$

so the k entries of Y are i.i.d. standard normal, and

$$S_k = \mathbf{z}^\top \Sigma_k^{-1} \mathbf{z} = \mathbf{z}^\top (L^\top)^{-1} L^{-1} \mathbf{z} = \|Y\|^2 \sim \chi_k^2.$$

Incremental update. Write Σ_k in block form,

$$\Sigma_k = \begin{pmatrix} \Sigma_{k-1} & \Sigma_{1:k-1, k} \\ \Sigma_{1:k-1, k}^\top & \Sigma_{k, k} \end{pmatrix},$$

where Σ_{k-1} , the leading $(k-1) \times (k-1)$ block, was already factorized as $\Sigma_{k-1} = L_{k-1} L_{k-1}^\top$ at the previous step. Since L_k is lower triangular, its upper-right block must vanish, so $L_k = \begin{pmatrix} A & 0 \\ \ell^\top & L_{k, k} \end{pmatrix}$

for some $A, \ell, L_{k, k}$. Expanding $L_k L_k^\top$ and matching blocks against Σ_k gives three equations: $AA^\top = \Sigma_{k-1}$, $A\ell = \Sigma_{1:k-1, k}$, and $\ell^\top \ell + L_{k, k}^2 = \Sigma_{k, k}$.

The first equation only states that A is *some* Cholesky factor of Σ_{k-1} ; but A , as the leading block of a lower-triangular matrix with strictly positive diagonal, is itself lower triangular with strictly positive diagonal, and that factor is unique. So $A = L_{k-1}$ exactly: the leading block of L_k is the identical matrix already computed at step $k-1$, not merely equal in value. The second equation, $L_{k-1} \ell = \Sigma_{1:k-1, k}$, is then solved for ℓ by forward substitution at $O(k^2)$ cost. The third gives $L_{k, k} = \sqrt{\Sigma_{k, k} - \ell^\top \ell}$, well-defined because $\Sigma_{k, k} - \ell^\top \ell$ is the Schur complement of Σ_{k-1} in Σ_k , which is positive whenever Σ_k is positive definite.

The solution vector updates by the same argument. Writing $Y_k = L_k^{-1} \mathbf{z}^{(k)} = (w_{1:k-1}^\top, \alpha_k)^\top$, the block form of L_k turns $L_k Y_k = \mathbf{z}^{(k)}$ into $L_{k-1} w_{1:k-1} = \mathbf{z}^{(k-1)}$ for the leading rows – exactly the system defining Y_{k-1} – so uniqueness again forces $w_{1:k-1} = Y_{k-1}$ exactly. The final row then gives $\alpha_k = (z^{(k)} - \ell^\top Y_{k-1}) / L_{k, k}$.

Preserving the null distribution. By the base case, $Y_k \sim \mathcal{N}(0, I_k)$ under H_0 for every k , so each new entry α_k is standard normal, independent of $\alpha_1, \dots, \alpha_{k-1}$. Since $S_k = \sum_{i \leq k} \alpha_i^2 = S_{k-1} + \alpha_k^2$, this is exactly the running sum of k independent squared standard normals required for $S_k \sim \chi_k^2$ to hold at every prefix length – matching the base case above without a fresh factorization to re-establish it. $\square$

D Progressive scaling and runtimes

Table 4 summarises the five stages the pipeline was progressively validated and scaled through, from a tractable proof of concept to the full dataset reported in Section 2.5. Stages 1 and 2 enumerate subsets exhaustively rather than sampling, since $K \leq 10$ keeps the combinatorial space tractable; random sampling begins at Stage 3, where $K = 100$ first makes exhaustive enumeration infeasible ($\approx 1.3 \times 10^{30}$ subsets). The two stages also predate the LDSC intercept matrix, introduced properly at Stage 3 (Section 2.2): Stage 1 uses the empirical phenotypic correlation directly, later shown to be biased upward by genetic signal (Appendix A), and Stage 2 uses a JASS-inspired proxy restricted to weak-signal SNPs as a cheaper approximation to LDSC.

Stage	SNPs	K	Covariance estimator	Subsets (N_s)	Cores	RAM allocated	Runtime
1. Pilot	5,249	5	Naive correlation	31 (exhaustive)	–	–	<1s
2. Scaling up	100,000	10	JASS-inspired	1,023 (exhaustive)	–	–	≈ 13 s
3. $1M \times 100$	999,991	100	LDSC intercept (block)	10,000	16	128 GB	1h 41m
4. Halfway	4,999,941	500	LDSC intercept (block)	10,240	32	150 GB	4h 15m
5. Full scale	9,611,261	1,010	LDSC intercept (full)	10,000	64	180 GB*	5h 17m

Table 4: Dataset size, covariance estimator, and compute resources at each stage of progressive scaling in the random subset sampling.

*Peak memory usage is 100 GB